Math 526, S12 Name: Answer

Quiz 6 Section: SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$.

(a) Reduce $\mathbf{A}$ to the echelon form $\mathbf{U}$ by Gaussian elimination.

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ -2 & -6 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

(b) Which are the free variables and which are the pivot variables?

Pivot variables: x_1, x_3

free variables: x_2, x_4

(c) Find the null space of $\mathbf{A}$, $N(\mathbf{A})$.

$AX = 0 \Leftrightarrow UX = 0$

$x_1 + 3x_2 + x_3 + 2x_4 = 0$

$2x_3 + 4x_4 = 0$

i) Set $x_2 = 1, x_4 = 0$, we get $x_1 = -3, x_3 = 0$

$\Rightarrow S_1 = (-3, 1, 0, 0)$

ii) Set $x_2 = 0, x_4 = 1$, we get $x_1 = 0, x_3 = -2$

$\Rightarrow S_2 = (0, 0, -2, 1)$

Thus $N(A) = \left\{ c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$.